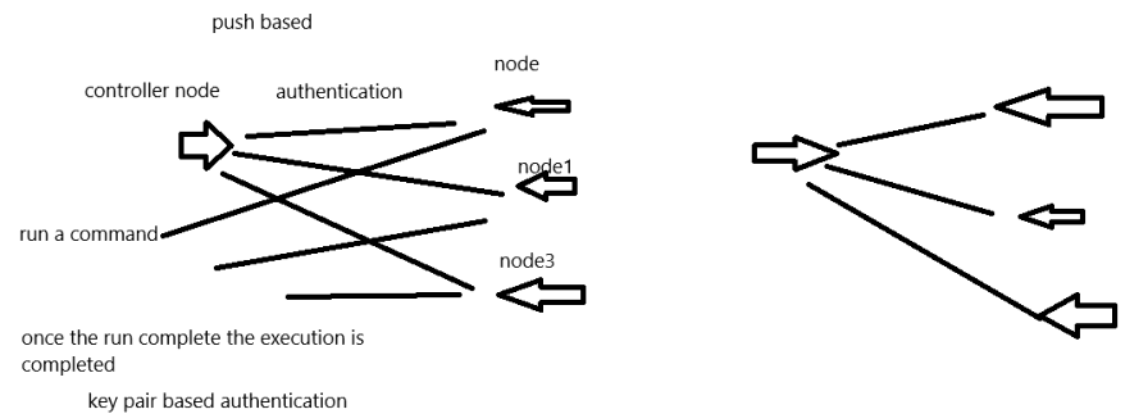
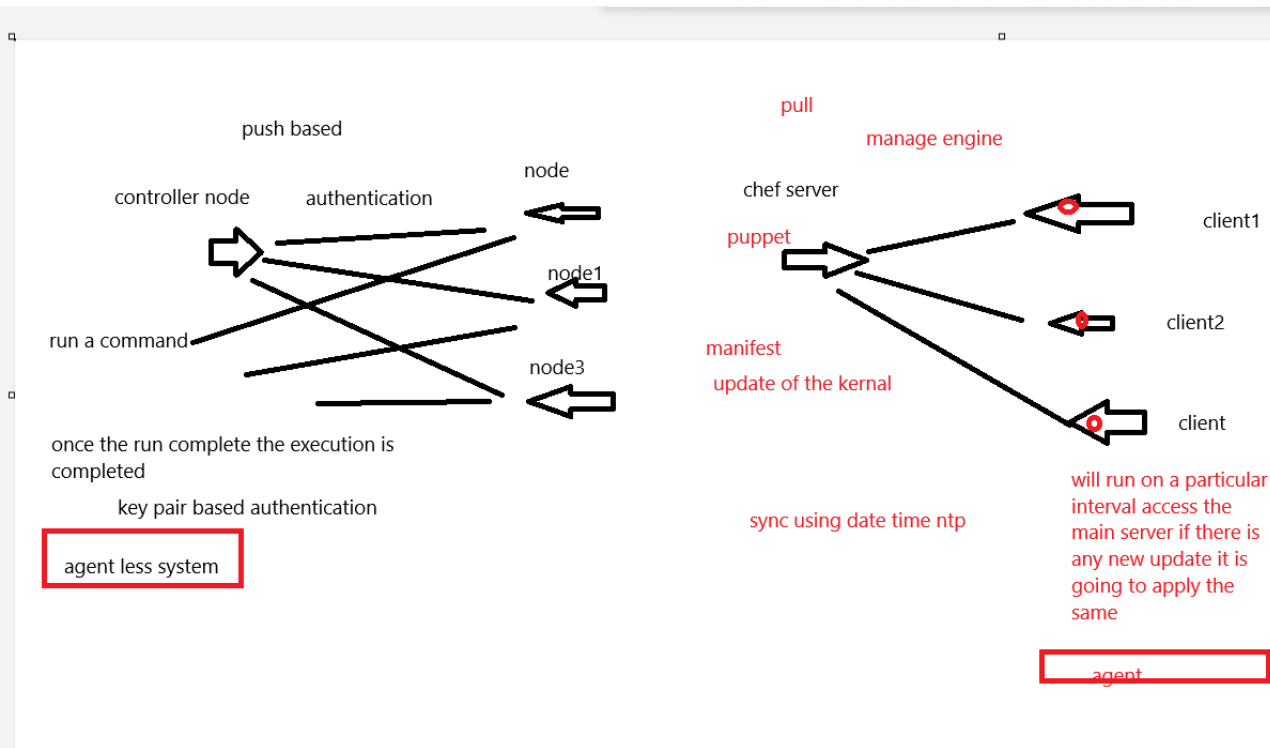




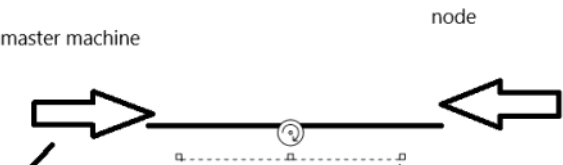
Screen clipping taken: 8/3/2026 10:21 AM

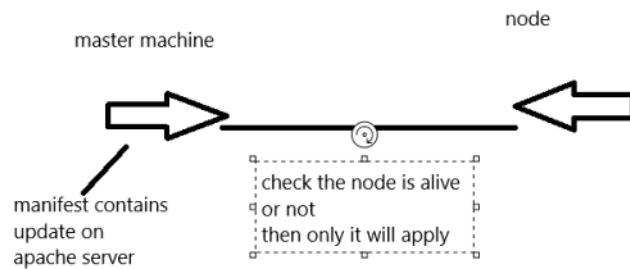


Screen clipping taken: 8/3/2026 10:25 AM

When should we use agent less or agent based system

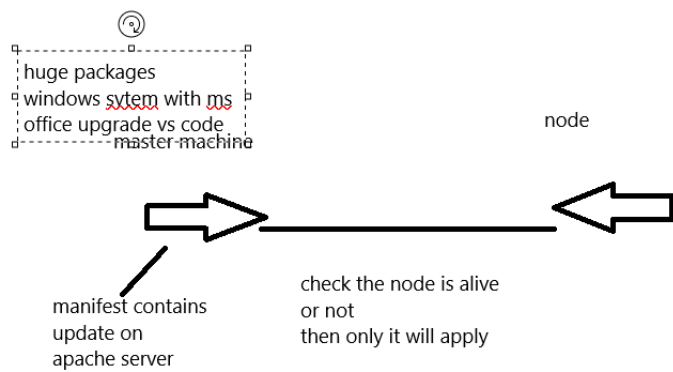
1. Software deployment on real time basis.





ansible node
apache.yml

Screen clipping taken: 8/3/2026 10:33 AM



ansible node
apache.yml

Screen clipping taken: 8/3/2026 10:35 AM

Agent time
Sometime we need agent and agentless both the system

Ansible

Monday, August 3, 2026 10:38 AM

Can we use ansible to provision resources in the cloud?

We can use ansible to do the same. Boto3 library

Terraform is quite easier to do it.

2. IAC we write it once and reuse it. Well secure architecture template. Security risk
3. Vulnerability scanning. In Jenkins pipeline we will put
4. At the end there is no vulnerability in the code then only it will be ready for deployment ansible-lint

Ansible-lint: this will evaluate your code against syntax issues, best practices, standard deprecation

Ansible sanity test: ansible-test which will run static code analysis

Checkov

Ansible security scanner

Synk

Sonarqube

I don't want some of the packages to need to be installed as a root account but run as a non-root user

Docker requires root privileges to install

You can run Docker using a non-root user

I am launching a database. Logging inside the data we have put plain text password. I want the playbook to be encrypted using Ansible Vault

Dynamic testing. Before your code goes into deployment.

We run some Kubernetes pod manifests; you need to ensure they run as a non-root user with privileges

Order

Monday, August 3, 2026 10:58 AM

Order is important when you define a task.

I want to install nginx web server in few of the system

1. Update the system
2. Upgrade
3. Download the package
4. Install it nginx
5. Check
6. Test using ansible handler curl localhost 200 successful

Ansible all mynginx.yml

But there is no update what will happen run it and check if there is no update it is going to skip it.

Yaml

Monday, August 3, 2026 11:04 AM

What is yaml. To write an playbook.

Terraform build using go

Ansible build using python: using python whatever task we do using ansible we can do the same using python.

Easy to understand.

Do you think you need to build your own module for ansible.

Nrsc: they have all the application build using c language. Memory leak. Created problem in power processor from IBM.

We build an python include in ansible . Alert patch those application.

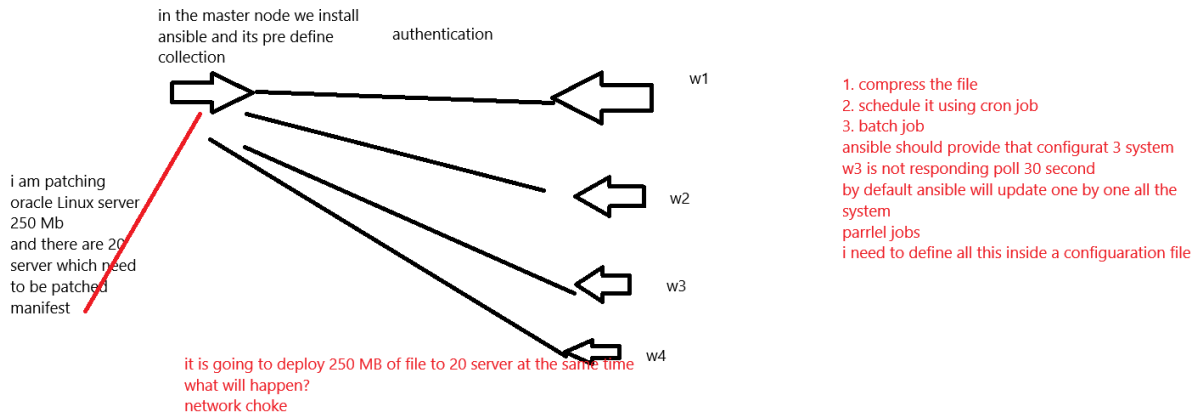
Most of things

Ansible module

Master node

Monday, August 3, 2026 11:14 AM

All the collection are installed only on master node



Screen clipping taken: 8/3/2026 12:13 PM

1. Project-
2. Ansible.cfg: this will control
3. Inventory : Ip address password or key pair auth grouping. Dynamic inventory
4. Yaml file-manifest file for ansible

Inventory

Monday, August 3, 2026 2:41 PM

How to manage the inventory

Authentication you need to do it by default between master and worker node

If your

So we learn about different ways to manage inventory

Using -I passing the inventory file from any directory or want to run all the inventory from a directory

Ansible web -m ping

Ansible-play web test.yml

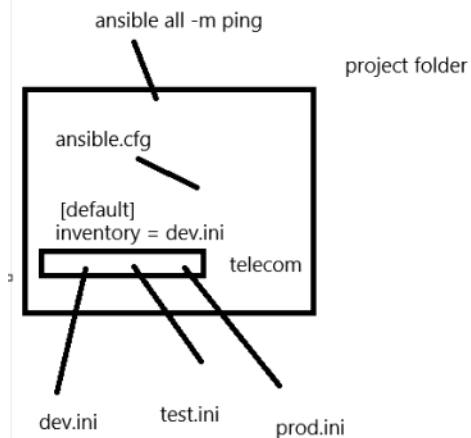
They will first look into ansible.cfg file inside that it will look for all default configuration

```
ansible.cfg
default
inventory = dev.ini
```

```
ansible -i test/prod.ini -m ping
```

overwriting the default ansible.cfg file

Screen clipping taken: 8/3/2026 3:37 PM

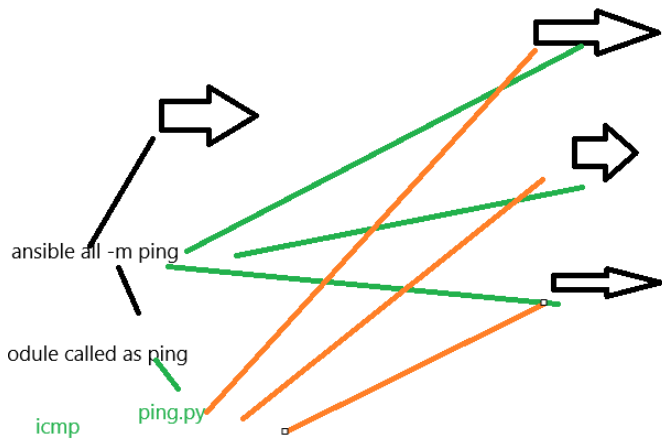


ansible -i ./telecom/prod.ini -m ping this will overright the default inventory path and run only prod.ini file which contains prod server inventory

ansible -i ./telecom/test.ini ./telecom/prod.ini -m ping

ansible -i ./telecom -m ping

Screen clipping taken: 8/3/2026 3:44 PM



Ansible is using ping module but between the master and worker node the connection we have is using ssh

Screen clipping taken: 8/3/2026 3:55 PM

Ansible ad hoc command

Monday, August 3, 2026 4:18 PM

Like I want to do some system admin task pathing or want to add a new user service to be restart trouble shooting.

Recap

Tuesday, August 4, 2026 9:48 AM

IAC using ansible is that . Frequent update patching.

As an configuration management work fine

Ansible master node can be installed only on linux system

On windows WSL.

Ansible package is available as a binary

We have look into an architecture where we create a project folder

Inside the project folder we have added a file called as ansible.cfg

Default parameter to run your ansible script

1. Inventory the inventory file extension can be .ini or yaml
2. Inside the inventory file we have define hostname and ip of the system
3. Ansible username or portnumber by default ansible it will look for which port 22 2222
4. Between the master and worker node the configuration work only if we do ssh key pair authentication

Ansible adhoc command

This command is used for quick fixes

This setup

Setup module

Tuesday, August 4, 2026 10:02 AM

It is collecting your system inventory

By default your run any ansible command it will first run the setup module. So that it gather your system information

Ansible will have the current state then only it can go for desired state fact checking

To get the current state what type of authentication is required in your system

Playbook

Tuesday, August 4, 2026 10:12 AM

Ad hoc command can do single task

If you want to do multiple task

I want to setup an nginx server apache or an mysql db or an tomcat server

1 update the system

Upgrade

Install the package

Check which os is running according to that the command will execute for installation

Amazon linux yum ubuntu apt

Enable the service

Put in the startup

Start the service

Check the url of the service

I need to create a manifest file inside that I need to provide all this steps

Ansible is using python

Using yaml

I want to install apache

1. Update my system
2. Upgrade my system
3. Custom configuration file for apache so that my apache server run on port 81
4. Install apache
5. Start the service
6. Enable the service

Container tech docker k8s everything run on yaml

Nginx playbook

Tuesday, August 4, 2026 11:23 AM

We need to write it using yaml file so we will use visual studio code

This yaml file should execute in two different os amazon linux and ubuntu linux

Yum apt

Before it can install the package we need to define which os it is

Jinja2 in python

How to call an variable in python

Which use jinja2 template

It start with {{nameofthevariable}}

Handler in ansible

Tuesday, August 4, 2026 2:11 PM

Whenever I run my play book it will execute all the task.

Every time the playbook runs nginx is restart even if the configuration file did not change

Using ansible we deploy the resource and handler has restarted the services

After that I need to do verification that my webpage is running or not.

Handler should verify the website show some website responses

How to manage container using ansible

Tuesday, August 4, 2026 4:18 PM

1. Install docker
2. Start and enable docker service
3. Pull an ubuntu image
4. Runs an ubuntu container
5. Login inside the container and run some command

Ansible can execute command inside a container but it cannot keep an interactive docker exec -it session open because play books are non interactive.

If you want to do continuous interaction you have to use the docker command

Docker exec -it nameofthecontainer bash

We will build a custom image

Spin up some container

And get the container ip address

Recap

Wednesday, August 5, 2026 9:36 AM

We start righting the playbook

1. Define the inventory where the play will run
2. We define variable to be used inside the playbook
3. Task
4. In task we have define list of items

What is difference did you see between dictionary and list

If a task fail we need to manage the failure gracefully without stopping the play

Dictionary

Wednesday, August 5, 2026 10:28 AM

We can use dictionary to create a playbook using ansible.

I want to create a user

I need to provide username, group shell

When I am create a user also want to create password

Docker project

Wednesday, August 5, 2026

11:26 AM

1. We install docker in both the machine and verify the same that docker is installed or not
 2. I want that docker daemon to be run by the normal user
 3. Ubuntu
 4. Amazon linux it should be run using ec2-user
 5. I want to build a container image
 6. Docker file application I want to get the docker file and application file from an version control system
 7. Build the image
 8. Show the image
 9. Build two container out of those image
 10. Get information about the container
 11. Upload the image in a docker hub username and password which we will encrypt using vault
- This is like an full fledge container cicd pipline using ansible

Command: `df -h uptime`

Every time I run the play book it will rerun it if there is change in code or not it will run it any how

We loose the idepotenc

Yes: after we build the image you can authenticate in docker hub push the image. This is recommended

1. Build the image
2. Login to `hub.docker.com` : vault to login to the `hub.docker.com`
3. Tag the image from your docker hub repo
4. Push the image
5. Finally we want to run the container from remote repo.

Roles

Wednesday, August 5, 2026 4:21 PM

In this playbook we have added everything. Which is not the right way to build your playbook
Any infra as code and want to use the same in production it need to well organized.

Inside my play book I have around 200 lines of task

50 variables

10 handles

Templates

Files

Configuration

Then it will have some dependency on other playbook

Like I want to install tomcat on a linux machine.

Java

Top of the same I need to add java framework.

1. Install docker
2. Start docker
3. Configure docker
4. Build image
5. Run image in local
6. Build image as per hub
7. Upload the image
8. Push the image to hub
9. Create user
10. Creating group
11. Add user to the group

Proper structure so that we can easily understand what is this play.

Genie make a wish.

You need to design your playbook like that so that ansible playbook it will execute from top to bottom

I modify my app so only I want to run 6 7 8

So we need to create our playbook generalized whatever task we want to execute without running all the task if I can run only the required task

1. Using ansible roles
2. Using tags

Ansible deployment

Roles

Thursday, August 6, 2026 9:42 AM

Today we are going to learn about roles.

1. We have some common task like installing packages
2. Docker install docker verify configure
3. Application clone build it run verify it
4. Dockerhub task will login push deploy

Using ansible I want to create an k8s cluster

Master node installation multiple master node etcd setup

You worker node setup

Cni plugging

Containerd or podman or crio container.

Nodesetup whenever a new node is created we can make that part of your cluster

Arrange all your task in a folder with an proper directory structure

You can create test cases for your roles

Common folder we are install git

- Verify git installed git --version docker --version
- Docker it is running
- Ansible_fact.service['docker.service'].state=running
- Ansible_built.stat path{build.path} it exist
- Curl localhost:8082 200
- Container using trivy
- Idempotence test changed=0

Dynamic inventory

06 August 2026 11:47

Till now we have created the inventory and use the same. Static inventory

##why you need dynamic inventory in cloud shutdown instance start agains

Retrive your list and target server at runtime from external source like aws azure gcp

Inside your master node controller node

1. You need to have an plugin which will communicate with your cloud server
2. Cloud based cli tool aws cli azure cli gcp cli
3. Master node need to communicate with whom cloud service provider aws access key and secret access key service principal
4. Against that particular user you need to have a role which allow you to ec2 describe instance tags region

Git and jenkins

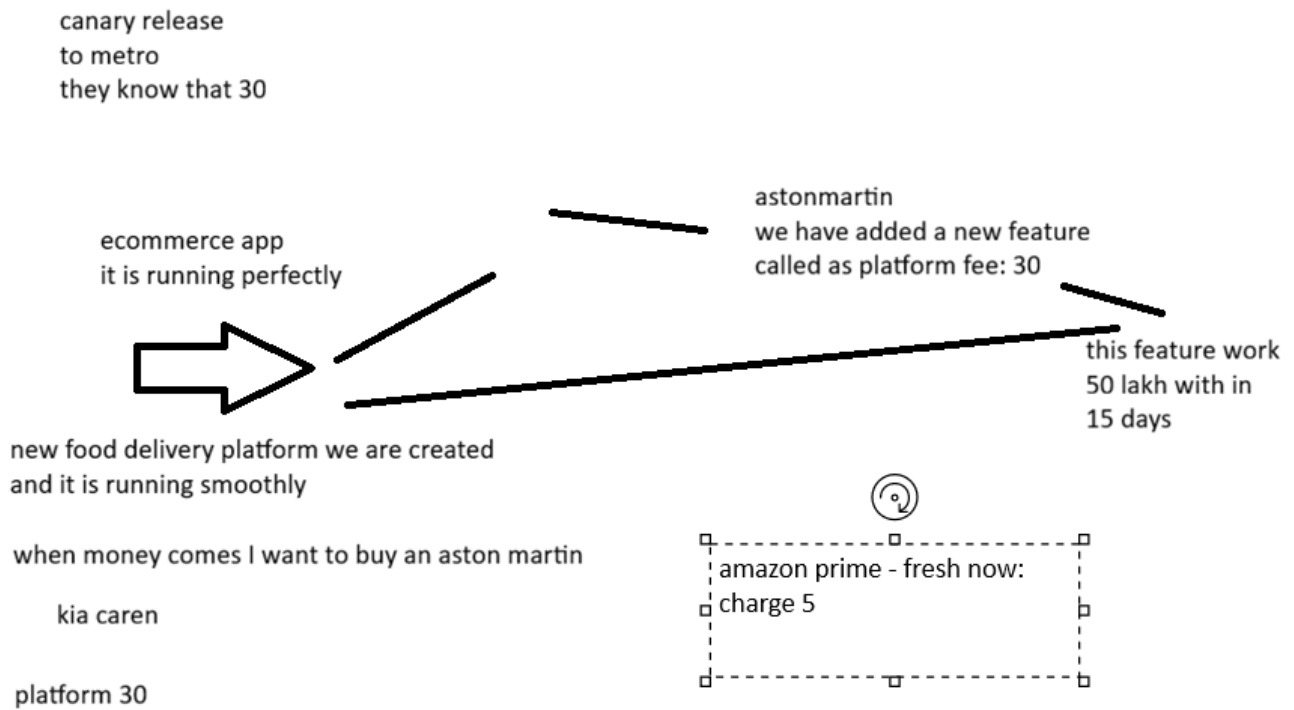
06 August 2026 14:09

We will learn about git and how to manage your ansible code using git

We will push the code to a remote repo

###to run it continuesly we will integrate the same with jenkins

So that next time when we make changes automatically the code will execute.



Screen clipping taken: 06-08-2026 14:17

Git exercise

06 August 2026 14:17

Using git we will manage our code upload in version control system

Authentication

Then we will create a new branch

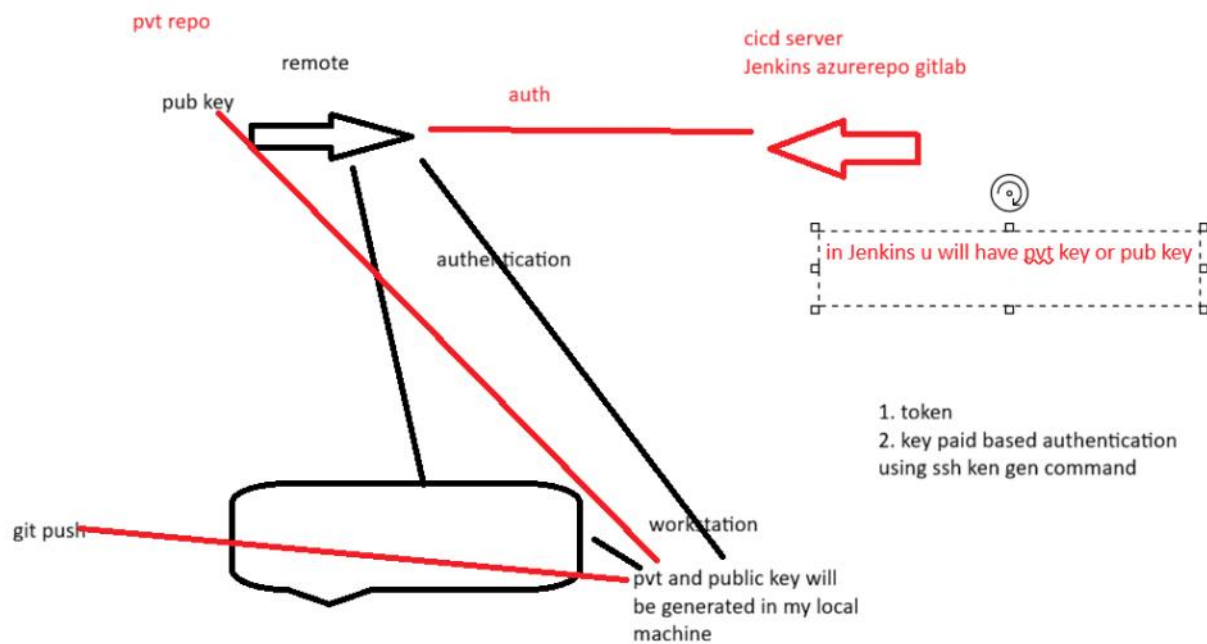
Add a new feature in that particular branch

All the changes are in local machine

We need to upload the same in remote repo github gitlab bitbucket azurepo

Downline the command to manage your code will be always git

I need to upload my code from local machine to the remote repo



Screen clipping taken: 06-08-2026 14:54

Jenkins

06 August 2026 16:10

Lets install jenkins

Jenkins pipeline parameter

Friday, August 7, 2026 10:03 AM

Discard old build: 30 a jar or war file created : after 30 days first in first out

Do not allow concurrent build: I push my code 1 time after few second I push my code again. Is in running state

The jenkins master server. Update the os kernal
Reboot jenkins will execute all the jobs again

There are multiple project which is running inside your jenkins server
I want to give priority to my project
You are doing what cpu ram

Preserve stash value: when you are executing a pipeline

1. Build : there is some file which is running
2. Test: some file which is in the running
3. Sonartest : running
4. Archival : running

When the pipeline is running in that time duration of the build it will stash your previous keep the steps in a temp file.

Once it is completed the stash is deleted.

Option {

preserveStash(buildCount) : 20 for the last 20 complete build preserve the stash

I push my code jenkins server at 1:00

I push it second time at 1:15

Throttling build: 60 min 2:15

Build after other project are build

What is the difference between build periodically poll scm

As both of them run using cron value scheduler

Build peridically: 15 min : if there is any change in the code or not it is going to build it

Poll scm: every 15 min it will check the code for any new changes if there is change it will build or skip.

You have created test cases which need to run every night check the ports number getting logs from application

Trigger build remotely

Acme.com/job/first-pipeline-code/build?token=TOKEN_NAME or /buildWithParameters?

token=test12

Custom pipeline

Friday, August 7, 2026 10:55 AM

We will create our own custom jenkins pipeline

Where we will clone an spring boot application

Build it

Unit test

Integration test

Checkstyle plugin

Finally integrate with sonarqube

Upload the same in nexus artifact

Containerized

Test

Friday, August 7, 2026 2:19 PM

Unit test

Lint testing

Sntax check

###all this are sast testing

We can use an tool called as sonarqube which is an testing framework. Single framework to do all the testing dashboard where you will find the whole testing.

We will integrate sonar with your jenkins pipeline.

Code coverage how much of your code is executed by your autmated test, dependecy test unit test integration testion

Sonar cube is not done the code coverage

Jacoco plugin to get coverage

Branch coverage which will check whether all decision path have been tested or not

Line coverage

Agent

Friday, August 7, 2026 4:27 PM